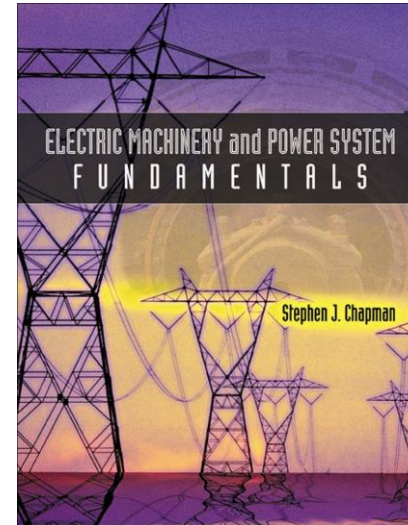


Energy Conversion

Fall 2023



ENERGY CONVERSION (Course Notes Version 2)

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Overview

1

Part I: Review of Fundamentals of Energy Conversion and Important Concepts

- Physics behind the Concept of Energy Conversion
- Electric Circuit Analysis
- Three-phase Circuit Analysis

2

Part II: Devices of Energy Conversion

- Transformers
- DC Machines
- AC Machines

3

Part III: Renewable Energy Resources

- Developments in Renewable / Alternative Energy technology
- New Energy Conversion Systems in Future Smart Grid
- Wind Energy
- Solar Energy

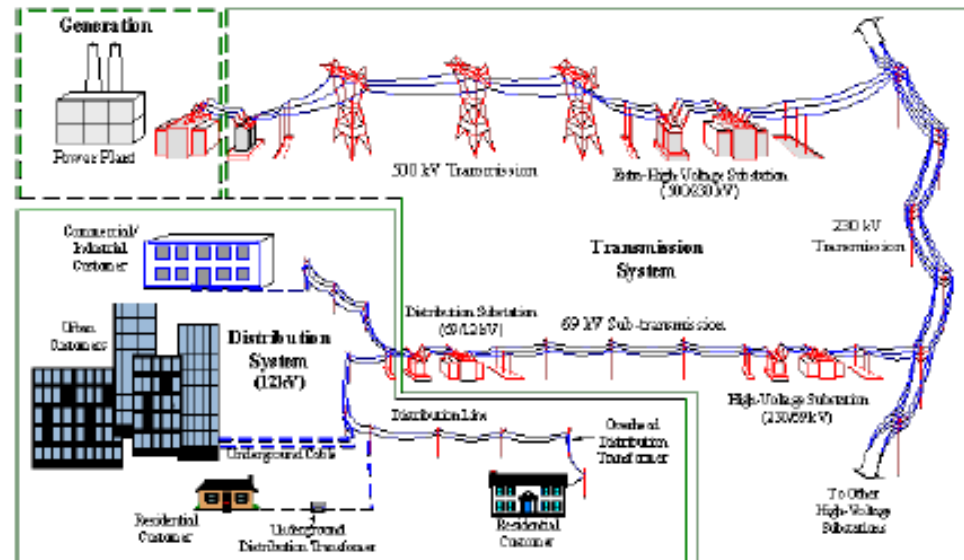
Chapter 1:

Overview of Energy and Power Industry from Various Aspect

► Structure of Power System:

An electrical power system is an interconnected network that delivers the generated power to the consumers. Generating stations, transmission lines and the distribution systems are the main components of an electric power system. Generating stations and a distribution system are connected through transmission lines, which also connect one power system (grid, area) to another. A distribution system connects all the loads in a particular area to the transmission lines.

- **Generation Systems**
- **Transmission Systems**
 - Transmission Lines
 - Substation
 - Substation Transformer
 - Surge Arrestor
 - Circuit Breakers
 - Sub-transmission Lines
- **Voltage levels**
- **Distribution Systems and Loads**





Chapter 1: Overview of Energy and Power Industry from Various Aspect

▶ Operation of Power System

▶ Power Delivery

Once generated, electricity travels across the regions from generating plants over high-voltage transmission lines to local transformers. The transformers reduce the voltage level making it suitable for local distribution. Electricity then travels over low-voltage distribution lines, owned by your local distribution provider, to your home or business. Power system has to be operated even when there is a moderate scale of failure. To ensure the system does not fail, power system stability analysis is carried out.

▶ Power System Stability

Power system stability refers to the ability of a system to return back to its steady state when subjected to a disturbance.

▶ Power System Failure

A power failure (also known as a power cut, power outage, power loss, or blackout) is the loss of the electric power to an area.

▶ Power System and Market Operation

1. It is completely interchangeable
2. It must be produced and used simultaneously since electricity storage is still expensive.

Due to these differentiating characteristics of electricity, Independent System Operators (ISOs) and Regional Transmission Organizations (RTOs) are responsible for keeping the power grid balanced between generation and load.

Chapter 2:

PHYSICS FUNDAMENTALS FOR ENERGY CONVERSION

▶ Electric and Magnetic Fields and their Interactions

- ▶ **Oersted's law:** Oersted's discovery states that when an electric current passes through a wire it creates a magnetic field around it.
- ▶ **Ampere's Law:** For any closed loop path, the sum of the length elements times the magnetic field in the direction of the length element is equal to the net electric current enclosed in the loop.

▶ Magnetization

▶ Magnetic Flux and Flux Density

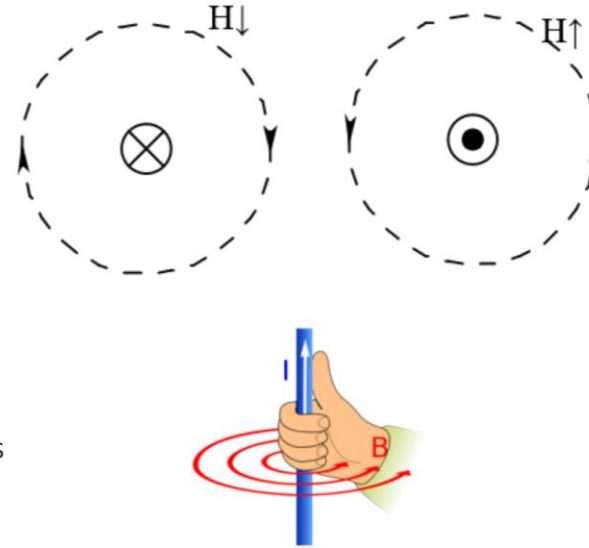
▶ Magnetic Materials and their characteristics

▶ Quantification (The Hysteresis Loop)

▶ Related Properties of Material

- ▶ **Permeability:** A property of a material that describes the ease with which a magnetic flux is established in the component.
- ▶ **Reluctance:** The opposition that a ferromagnetic material shows to the establishment of a magnetic field. Reluctance is analogous to the resistance in an electrical circuit.
- ▶ **Retentivity:** A measure of the residual flux density corresponding to the saturation induction of a magnetic material. In other words, it is a material's ability to retain a certain amount of residual magnetic field when the magnetizing force is removed after achieving saturation.
- ▶ **Residual Magnetism or Residual Flux:** It is the magnetic flux density that remains in a material even when the magnetizing force is reduced to zero.
- ▶ **Coercive Force:** The amount of reverse magnetic field which must be applied to a magnetic material to make the magnetic flux return to zero.

Chapter 2: PHYSICS FUNDAMENTALS FOR ENERGY CONVERSION



▶ Electromagnetism

- ▶ **Changing Current produces Changing Magnetic Field**, the magnetic field is dependent on the current in following ways:
 - ▶ 1. Change in current direction.
 - ▶ 2. Change in current magnitude.
 - ▶ 3. Change in both direction and magnitude.
 - ▶ 4. Change in density.
- ▶ **Faraday's law:** Faraday's law of induction describes a basic law of electromagnetism, involved in the working of transformers, inductors, and many forms of electrical generators. The law states: The induced electromotive force (emf) in any closed circuit is equal to the time rate of change of the magnetic flux through the circuit.
- ▶ **Lenz's Law:** The direction of the electromotive force is given by Lenz's law. The law states: An induced electromotive force (emf) always gives rise to a current whose magnetic field opposes the change in original magnetic flux.
- ▶ **Magnetic Circuits (Equivalent circuits)**

Chapter 3: REVIEW OF ELECTRICAL CIRCUITS FOR ENERGY CONVERSION

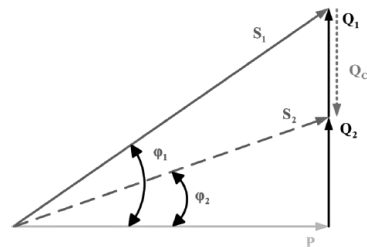
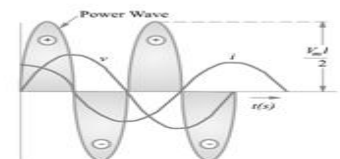
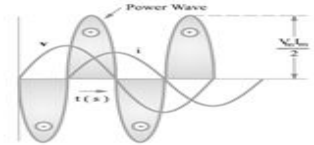
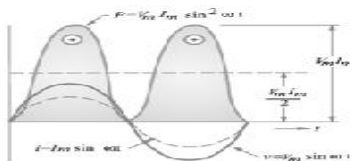
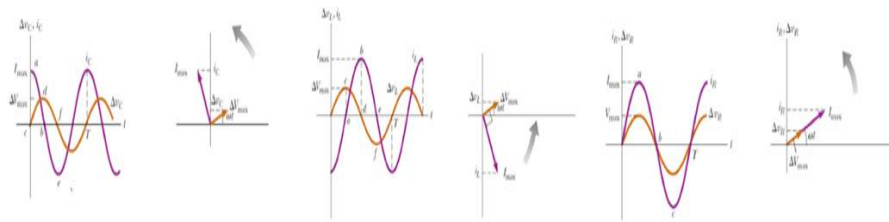


Figure 3.10 Power factor correction.

► Discussion of Important Concepts and Techniques

- Resistive
- Inductor
- Capacitor

► Complex Power in AC circuits

In AC circuits, the instantaneous values of the voltage, current and therefore power are constantly changing as a function of time therefore, the calculation of power in AC circuits is different to that of in DC circuits, but conceptually it is equal to the product of voltage (v) and current (i).

- Resistive
- Inductive/Capacitive:

► Power Factor

Power Factor is the ratio between the useful power (kW) to the total apparent power (kV A) that is supplied to the electrical equipment.

► Power Factor Correction Calculation

Chapter 4:

THREEPHASE AC CIRCUITS

- ▶ **Introduction to Three-phase Systems and Basic Concepts**
 - ▶ Phase Displacement in Single Phase Circuits (Leading, Lagging and In-phase)
 - ▶ Two phase and Three-phase Power Sources and Energy Systems
 - ▶ Symmetry and Balance Concepts in Three-phase Energy Systems
 - ▶ Phase Sequences (Positive, Negative and Zero Sequences)
- ▶ **Connections in Three-phase Systems**
 - ▶ Wye Connection and Neutral Voltage and Current
 - ▶ Delta Connection and Loop Voltage and Current
 - ▶ Difference between Ground Line and Neutral Line
- ▶ **Line and Phase Quantities in Three-phase Systems**
 - ▶ Line Quantities (voltages, currents and powers)
 - ▶ Phase Quantities (voltages, currents and powers)
 - ▶ Line Quantity-Phase Quantity Conversion
- ▶ **Single Phase Representation of Symmetrical Balanced Three-phase System**
 - ▶ Derivation of Equivalent Voltage Source
 - ▶ Derivation of Equivalent Load Impedance
- ▶ **Power in Three-phase Systems**

Chapter 5: TRANSFORMERS

- ▶ **Operating Principle of Transformer and Basic Concepts**
 - ▶ Operating Principle
 - ▶ Introduction to Basic Concepts with Ideal Transformer
- ▶ **Practical Transformers**
 - ▶ Things to Consider in Practical Transformers
 - ▶ Modeling of Practical Transformer
- ▶ **Analysis of Practical Transformer**
 - ▶ Conversion between the Primary and the Secondary
 - ▶ Approximate Equivalent Circuits
- ▶ **Three-Phase Transformers**
 - ▶ Three-Phase Transformer from three Single-Phase Transformers
 - ▶ Three-Phase Transformer Connections

Chapter 6:

DC MACHINES

- ▶ **Overview of History of DC Machines**
 - ▶ Inception of DC Machines
 - ▶ Traditional DC Machines
 - ▶ Modern DC Machines
- ▶ **Overview of Physics of Operation of Transitional DC Machines**
 - ▶ Generator Actions
 - ▶ Motor Actions
- ▶ **Construction of DC Machines**
 - ▶ A Conceptual DC Machine
 - ▶ Key Features of DC Machine
 - ▶ Commutator and Brushes
 - ▶ Armature Reaction
- ▶ **Types of DC Machines:**
 - ▶ DC Generator
 - ▶ DC Motor

Chapter 6:

DC MACHINES

- ▶ **DC Generator**
 - ▶ Operating Principle
 - ▶ Equivalent Circuit Model of DC Generator
- ▶ **DC motor**
 - ▶ Operating Principle
 - ▶ Equivalent Circuit Model of DC Motor
- ▶ **Output Characteristics of DC Motor**
 - ▶ Angular Velocity and Output Torque Relation
 - ▶ Derivation of Output Characteristics, Two Char. Points, DC Motor Efficiency
- ▶ **Classification of DC Motors**
 - ▶ Separately-excited DC Motor
 - ▶ Series-excited DC Motor
 - ▶ Shunt-excited DC Motor

Chapter 7:

AC MACHINES

- ▶ **Three-phase Rotating Magnetic Field**
 - ▶ Setup
 - ▶ Mathematical Representation
 - ▶ Physical Perspective
- ▶ **Synchronous Machines**
 - ▶ Construction
 - ▶ Synchronous Generator
 - ▶ Operating Principle
 - ▶ Synchronous Speed and Number of Pole Pairs
- ▶ **Synchronous Motor**
 - ▶ Operating Principle
- ▶ **Asynchronous/Induction Machines**
 - ▶ Construction
 - ▶ Induction Motor
 - ▶ Operating Principle
 - ▶ Slip and Slip Ratio
 - ▶ Equivalent Circuit
 - ▶ Derivation of Equivalent Circuit
 - ▶ Analysis using Equivalent Circuit
 - ▶ Testing of Induction Motor
 - ▶ Induction Generator
- ▶ **Output Characteristics of Induction Motor**
 - ▶ Electric Power Output
 - ▶ Mechanical Torque Output

Questions

